Fakultät für Mathematik und Informatik
Societal Computing

UNIVERSITÄT DES SAARLANDES

Project Seminar
'Data Science and Artificial Intelligence'

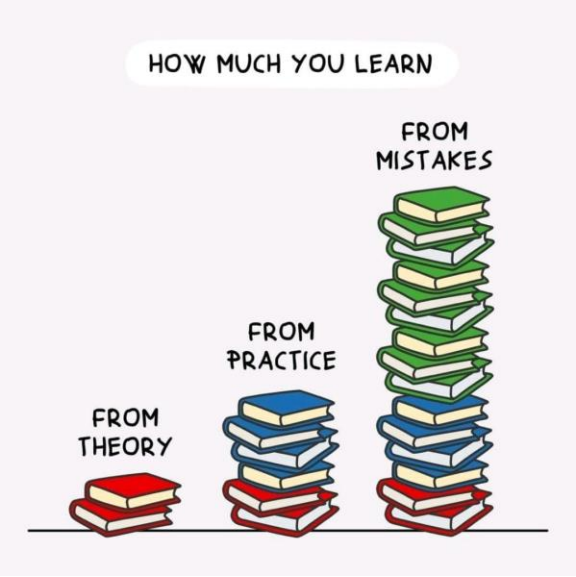
Kick-Off

SIC Saarland Informatics Campus

What is the DSAI Project Seminar for?

"In theory, there is no difference between theory and practice. In practice, there is." Yogi Berra (?)

https://www.espn.com/mlb/story/_/id/13722210/yogi-berra-quotes-celebrating-late-great-new-york-yankees-legend



https://www.reddit.com/r/GetMotivated/comments/znfwuf/image_how_much_you_learn_from_theory_practice_and/

DSAI is more than …
- Coding ML algorithms
- Loading data into a database
- Writing Python code

It also requires …
- Understanding client needs
- Thinking creatively about data
- Communicating well
- Working in a team

This (ungraded) project seminar is a safe environment to make mistakes – and to learn from them.

Team

Dr. Till Koebe

Prof. Dr. Ingmar Weber

No Grey Area!

(Sexualized) Discrimination and violence are not tolerated at university - neither in presence nor digitally!

Overstepping the bounds!

Unwanted and unwarranted behaviour, (physical) contact, actions
Physical assault
Stalking
Distribution of sexist content
Sexualized use of language
...

What can I do?

Pay attention and report things you notice!
Confront the person directly with their behaviour!
Seek help and draw attention to the assault out loud!
You find support at:

Gender Equality Office of UdS
Phone: 0681 - 302 4814
E-Mail: gleichstellung@uni-saarland.de

Organizational

- For DSAI Bachelor students only (already enrolled)
- At least one of the following courses must be passed
  - Elements of Data Science and Artificial Intelligence
  - Elements of Machine Learning
  - Programming I or II
  - or comparable
- 9 credits, not graded
- Applied group projects (2 students) with supervisors from different departments
- All important information on CMS: https://cms.sic.saarland/dsaiproject26/

Important Dates (all in CMS)

- 15.4. - Kickoff
- 20.4. - Register for seminar on CMS
- 20.4. - Register group on CMS
- 20.4. - Groups sends email containing your three favorite topics in ranked preference order
- 12.5. - Register for seminar on LSF
- 15.5. - Submit description of expected key figure via CMS (more details later)
- Teams meet regularly with your supervisors
- 17.6. - Poster submission via CMS (DIN A0, portrait, English preferred, printing will be done by us)
- 24.6. - Poster session: 10-12h at E1.3 HS 003
- 31.7. - Final report & code submission via CMS (report: 5 pages excl. references, English preferred, pdf)
- End of August – We add pass/fail to CMS/LSF
- All registrations and submissions are due end of day (that is 23:59h German time)

Notes on AI and AI Usage

- Our suggestion:
  - free to use AI assistants at any stage and any extent
  - we require you to add a section called "Notes on AI use". It should contain:
    - a) a summary how much and for what you used it
    - b) a critical reflection of your use of AI in your project:
      - What worked well?
      - What did not work well?
      - What would you do differently in your next project?
    - about a page long and written in flow text
  - **BUT: Donor's decision and students must stick to their donor's rules**

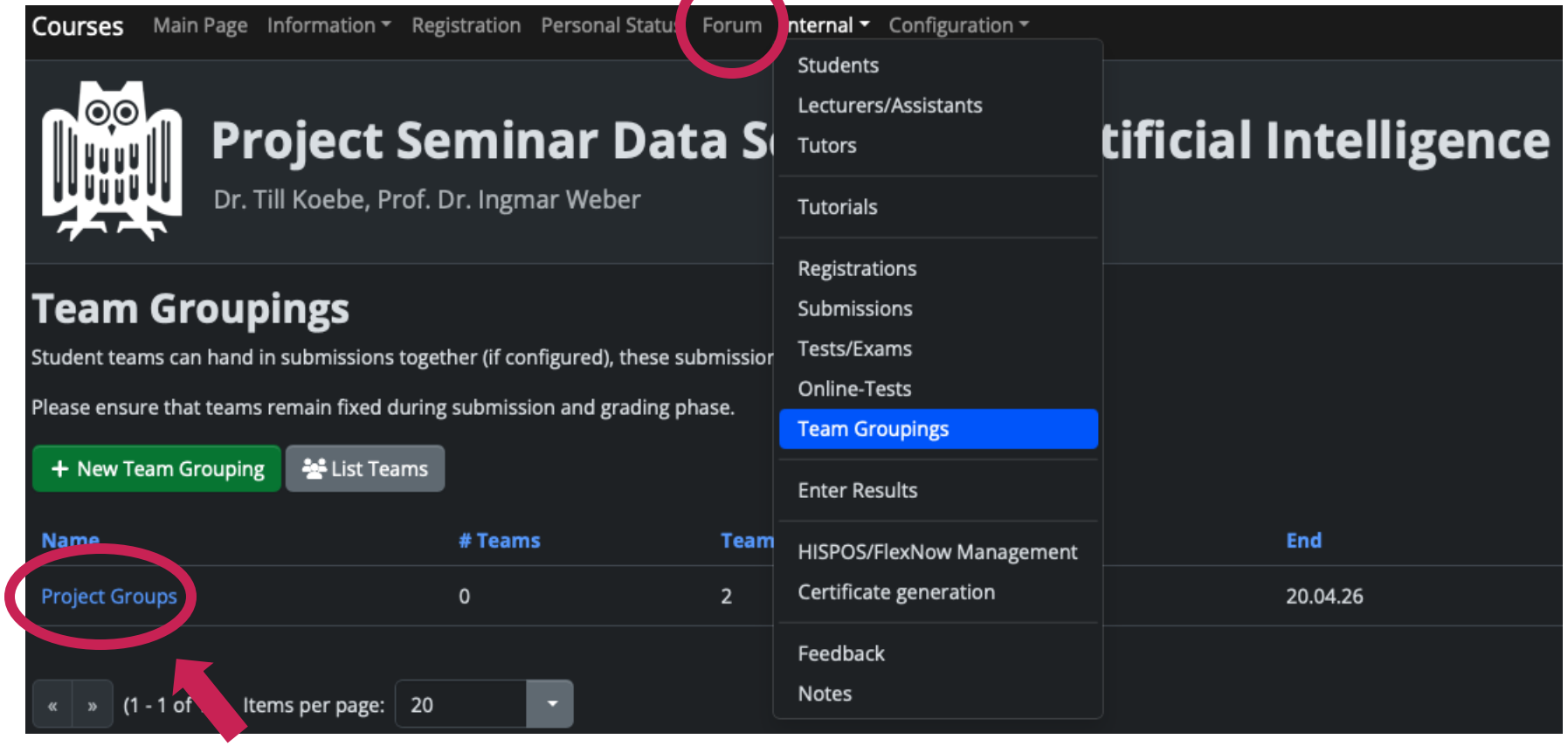
Further suggestions

- Start early!
- Note who contributed how to the project
- Contact us for administrative points
- Contact your project donor for topic-related points
- Whi

How to find/add a group

UNIVERSITÄT DES SAARLANDES
Fakultät für Mathematik und Informatik
Societal Computing

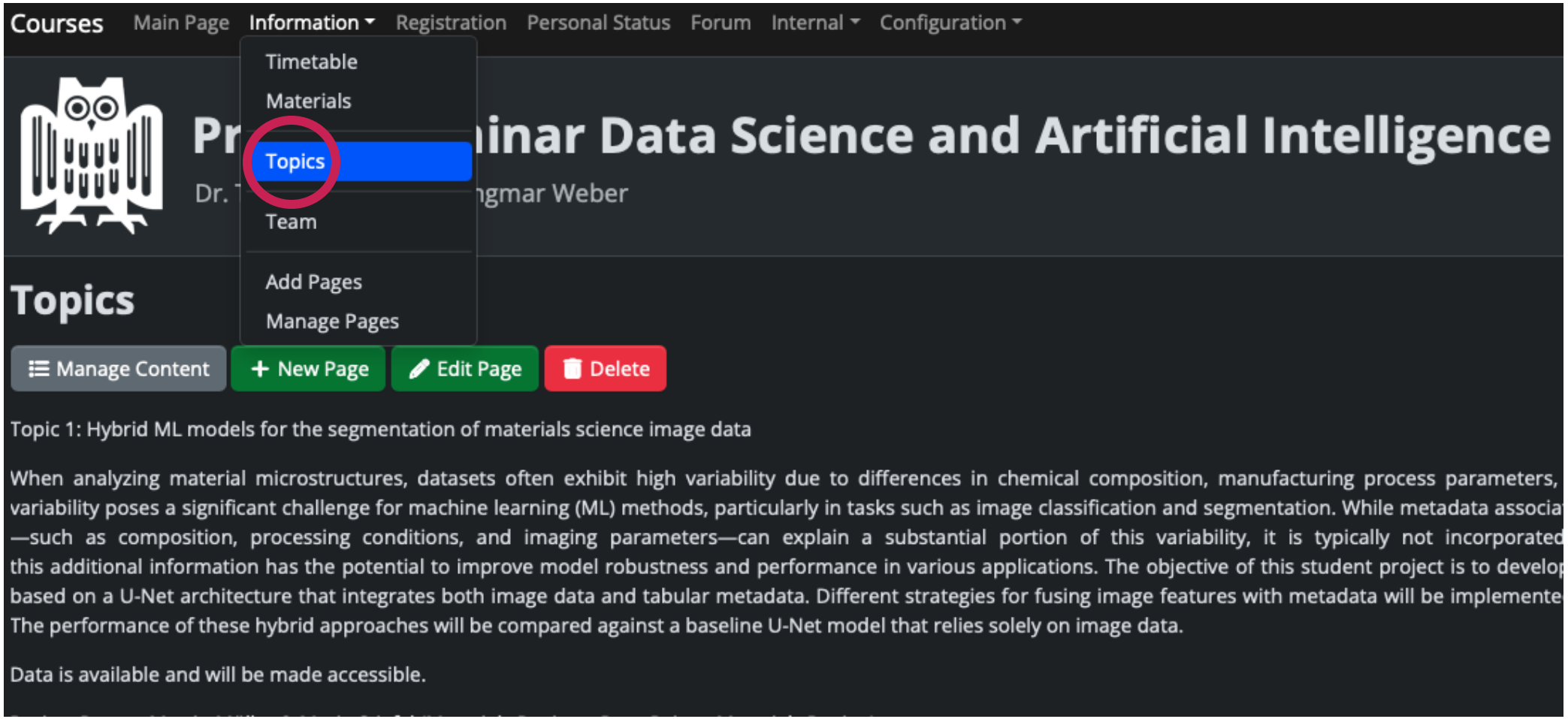
1. **Find group members!**



2. **Create a new team in 'Project Groups'**

3. **send one email per group with your group's top 3 topics to tkoebe@cs.uni-saarland.de**

Where to find the list of projects



Milestone I: Expected Key Figure due by May 15

- What is the key finding you want to communicate?
- What is on the x-axis?
- What is on the y-axis?
- What does the legend show?
- What is the caption?
- What data processing is needed?

-> Describe it in detail on 1 page.

-> Build it with mockup data!

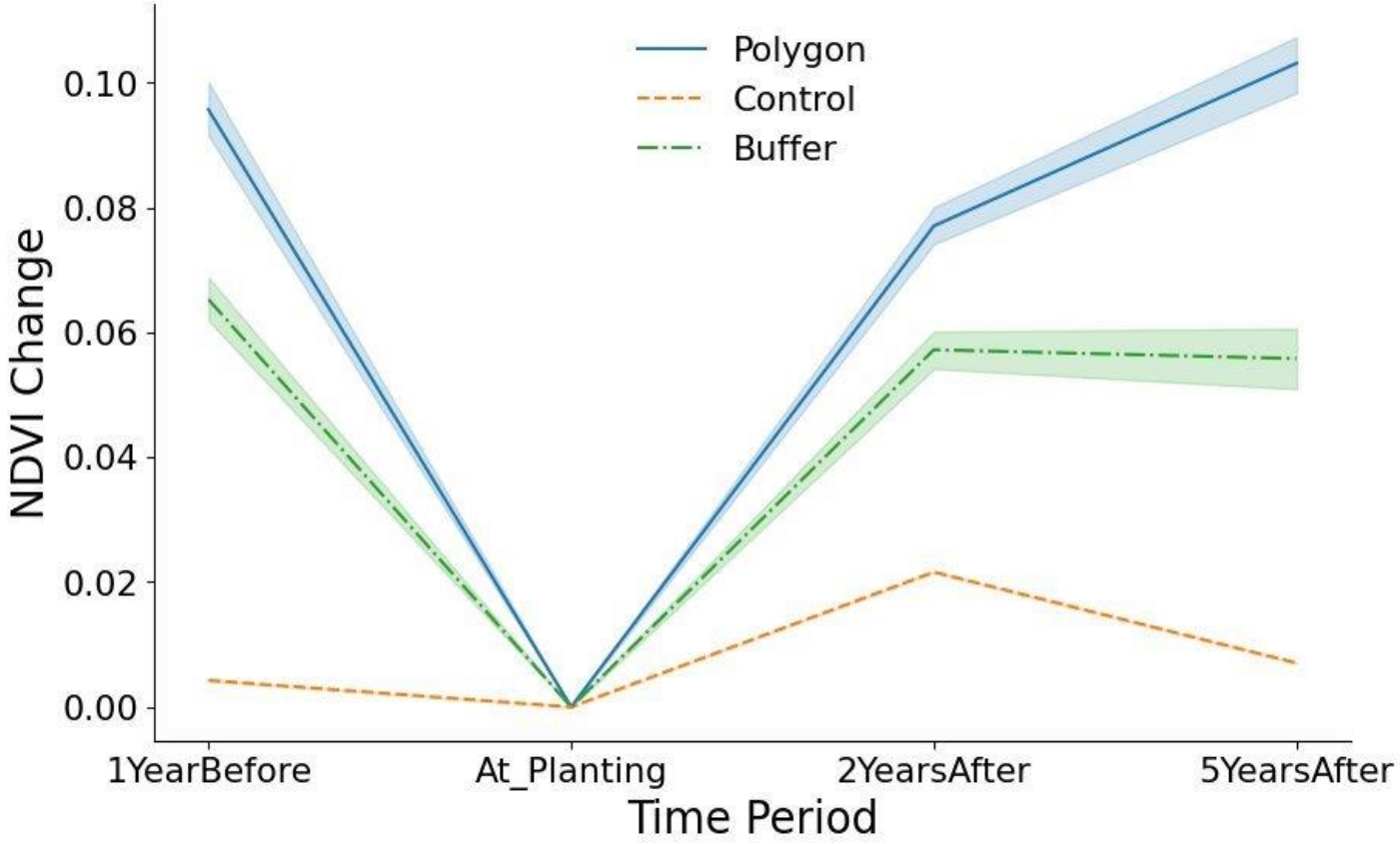


Figure 1: Changes in the normalized difference vegetation index (NDVI) at reforestation sites at different points in time around the planting date, by site type.

What will happen today in the remaining session

- Supervisors introduce themselves and their proposed projects
- Supervisors are:
  - Client (domain expert): Provides the problem (and ideally the data)
  - DSAI expert: Supports group to come up with a satisfactory solution
- We just do the organizational part

List of projects

#	Topic name	Presenter
1	Hybrid ML models for the segmentation of materials science image data	Marie Stiefel
2	Enhancing Dataset Diversity in Microscopy Images via Generative Augmentation	Björn-Ivo Bachmann
3	The impact of recent gas price surges and policy responses on mobility patterns in Germany	Jan Piontek
4	Best of both worlds? Integrating Computational Social Science and Political Survey Research	Philipp König
5	Between “L-takes” and “Löwengebrüll”: Mapping Ideological Structures in the Network of German Streamers and Influencers	Philipp König
6	Microtentacle-positive cells detection -- A quantification	Enrique Colina Araujo
7	Cost-Aware Machine Learning for Accelerated Drug Discovery	Michael Backenköhler
8	Investigating Non-i.i.d. Effects in Tennis	Luis Holzhauer
9	Quantify the relative role of skill and chance in determining match outcome of badminton and table tennis competitions	Luis Holzhauer
10	Surface-Specific Prediction of Serve and Return Performance in Tennis	Abdelrahman Abdelsamad
11	Multi-Aspect Network Performance Analysis of Multi-Parameter Measurements in a 5G campus network	Marlene Böhmer
12	Auditing platform algorithms using the pro-choice / pro-life posts on Instagram	Brahmani Nutakki
13	Quantum Machine Learning for Small Classification Tasks	Maximilian Kiefer
14	Tensor-Network-Based Adaptation Methods for Efficient AI Models	Maximilian Kiefer
15	Analysing open survey questions - How computer science can help to use their full potential	Martin Ulrich
16	Investigating Inequality Arguments from Historic Newspapers	Martin Schröder